Motor temperature: test and calibration plan

Inokim OXO on a VESC Labs Duet · rear 2500 W hub (CAN 48), front 1500 W hub (CAN 49) · script `lisp/dash_esc_lights.lisp` as installed on 2026-11-07

1. What is being calibrated

Neither hub has a temperature sensor, so the controller script estimates the temperatures. Three parts work together:

- **Heat model, both motors, always running.** Two nodes per motor: winding + stator (heated by copper loss 1.5·R(T)·(iq²+id²), R rising 0.393 %/°C, plus iron loss growing with speed^1.5) and shell + magnets (cooling to ambient better the faster the wheel turns). Ambient is the lowest controller FET temperature since power-on.
- **Rear resistance measurement at stops.** Standing still, throttle released, brake held for 2 s, at most once a minute: 25 A through the stationary rear motor for about 0.3 s. The winding temperature follows from the resistance rise against a cold reference. Rear only (motor 2 cannot be selected safely from the script).
- **Derating.** Above 100 °C winding or 80 °C shell the drive-mode current is reduced linearly, down to 30 % at 125 °C / 100 °C. It needs the display connected (it acts through the Dash ESC drive profile).

The model constants are first estimates. The plan measures the real rear winding temperature on a cold motor, during and after a hard climb, and fits the constants to those measurements.

Constants to be fitted (rear; front is scaled from the rear)

Name	Meaning	Start value	Fit range
`therm-cw1`	Winding + stator heat capacity	900 J/K	200–5000 J/K
`therm-cs1`	Shell + magnets heat capacity	1600 J/K	300–8000 J/K
`therm-rws1`	Thermal resistance winding → shell	0.25 K/W	0.03–2 K/W
`therm-g0-1`	Shell → air conductance, standing	1.0 W/K	0.1–10 W/K
`therm-g1-1`	Extra shell → air conductance per km/h	0.08 W/K	0–0.5 W/K
`therm-iron1`	Iron loss at 55 km/h	25 W	0–200 W
`therm-r0` / `therm-t0`	Cold reference resistance and its temperature (eeprom 100/101)	56.4 mΩ / 25 °C (motor detection)	set by `(therm-cal)`

2. Safety and how the work is split

Wheels-in-the-air rule. Every command that turns a motor or pushes current through it (`(therm-test)`, `(therm-cal)`, `(therm-cooldown …)`, any spin test) is sent by Claude only after Felix confirms in chat that the named wheel is off the ground or the brake is held, and nobody is near the wheels. Each measurement gives a small jerk as the rotor pulls to the nearest pole.

- **Hard temperature limit for the test:** stop climbing as soon as a measured rear winding temperature (`Wind M1 meas`) passes 110 °C, or the model passes 115 °C. Ride gently home; derating already starts at 100 °C.
- **Felix** does everything physical: parking, lifting the wheel, holding the brake, riding, the optional IR readings.
- **Claude** sends the REPL commands, pulls the logs, runs the fit, edits the script defaults in GitHub, uploads and verifies the script, all through `vesc_cli` on the laptop (runner window open, scooter on Wi-Fi).
- Values set with `setq` in the REPL are live only until the controller reboots or powers off. Anything permanent goes into the script defaults in GitHub and is uploaded.

3. Before starting

Item	Status / how to check
Script with heat model, stop measurement, cooldown command	Installed 2026-11-07, identical to GitHub (`check_sync.sh`).
Logging	Dash ESC logging on (Express). Log fields needed: `Wind M1`, `Shell M1`, `Loss M1`, `Therm amb`, `Wind M1 meas`, `R M1 meas`, `Therm derate`.
Laptop runner	`start_runner.bat` window open; Express reachable (192.168.1.155).
Room thermometer (recommended)	For the cold reference; any household thermometer.
IR thermometer + matt tape (optional)	For shell checks in phase 6.
Climb	The evening route home (140 m up), or a steep hill ridden 2–3 times.

4. Phases

Phases 0–2 can be done at home any time; phase 2 needs a motor that has sat for some hours. Phases 3–5 are one ride session (about 1.5 h including the 40 min cooldown). Phases 7–8 follow afterwards.

Phase 0	**Install check (no motor action)**
When	Any time, scooter on.
Setup	None.
Steps	Claude: `lisp repl --id 48 --expr "(list therm-r0 therm-t0 therm-meas-enable therm-meas-correct therm-amb)"`. Claude: pull the latest log and confirm the log fields listed in section 3 are present.
Pass	Reference values readable; `therm-meas-enable` = 1, `therm-meas-correct` = 0; all log fields present.
If not	Re-upload the script from GitHub and repeat.

Phase 1	**Repeatability of the resistance measurement**
When	At home, motor at room temperature or close.
Setup	Rear wheel off the ground (or rear brake held firmly). Confirm in chat.
Steps	Claude: `(therm-test)` five times, about 5 s apart (each: 25 A for 0.3 s). Claude records resistance, derived winding temperature and FET temperature for each.
Pass	Spread of the five resistances ≤ 1 mΩ (≈ 4 °C). Values near the detected 56.4 mΩ (corrected for temperature).
If not	Spread 1–2 mΩ: usable, average several points per stop. Above 2 mΩ: keep the stop measurement for logging only, leave `therm-meas-correct` at 0 and rely on the model; the fit is then not possible.

Phase 2	**Cold reference**
When	After the scooter has been parked indoors for at least 4 h (overnight is best), before riding.
Setup	Rear wheel off the ground or brake held. Note the room temperature. Confirm in chat.
Steps	Claude: check that the FET temperature is within about 2 °C of the room temperature (motor and controller have settled). Claude: `(therm-cal)`: averages 5 measurements and stores R0 and its temperature in eeprom 100/101 (survives power-off). Claude: read back `therm-r0` and `therm-t0` and note them for the fit.
Pass	R0 between 50 and 64 mΩ and consistent with phase 1; temperature equals the room temperature within 2 °C.
If not	If the FET temperature is far from the room temperature, wait longer and repeat. If R0 is implausible, repeat phase 1.

Phase 3	**Stop measurements in normal riding**
When	Start of the ride session.
Setup	Riding; at stops: standing still, throttle released, brake lever held for 2 s or more (the mechanical brake holds the wheel).
Steps	Felix: ride normally for 10 minutes, stopping 3–4 times with the brake held for 3 s; expect the small jerk. Claude (afterwards, from the log): `Wind M1 meas` changes at each stop; compare with `Wind M1`.
Pass	One measurement per qualifying stop (at most one a minute); measured and model within about 10 K at this stage.
If not	No measurements: check the brake switch reading (`ADC2` above 1.0 V while held) and the throttle at rest (`ADC1` below 0.55 V).

Phase 4	**Hot ride with dense measurements**
When	Same session.
Setup	Claude first sets (`setq therm-meas-interval 20.0`) (live until power-off).
Steps	Felix: ride the hardest climb available, at full throttle where safe. Stop every 2–3 minutes and hold the brake 3 s standing still. Aim: rear winding measured at 90–100 °C. Stop climbing above 110 °C measured (see section 2). Do not switch the scooter off between the climb and phase 5.
Pass	At least 10 stop measurements spread from cold to hot; peak measured ≥ 80 °C.
If not	If 80 °C is not reached, repeat the climb or carry it into a second session; the fit still works with a smaller range, with wider uncertainty.

Phase 5	**Cooldown curve**
When	Immediately at the top of the climb, scooter still on, logging running.
Setup	Scooter on its stand, rear wheel off the ground (or brake held throughout). Confirm in chat: this is a 40 min sequence of 40 measurements.
Steps	Claude: (`therm-cooldown 40 60.0`) (40 measurements, one a minute; stops by itself if the scooter moves or the throttle is touched). Felix: leave the scooter untouched; do not power it off.
Pass	40 points; the curve falls fast at first (winding to shell) and then slowly (shell to air). This shape is what separates the two nodes.
If not	If interrupted, restart with the remaining count; a gap of a few minutes is fine.

Phase 6	**Shell check with an IR thermometer (optional)**
When	During phase 5, every 10 min.
Setup	Matt tape on the hub shell (shiny metal reads wrong).
Steps	Felix: read the IR temperature on the tape and tell Claude the time and value.
Pass	Model `Shell M1` within about 8 K of the IR reading.
If not	Larger differences point at the shell constants; the fit can take the IR points into account as a check.

Phase 7	**Fit and update**
When	After the session.
Setup	None (no motor action).
Steps	Claude: `vesc_cli logs pull` for the session logs. Claude: `python3 tools/fit_thermal.py --r0 --t0 .` The fit prints new rear constants with uncertainties, scaled front constants, the RMS error before and after, the modelled peak, and a check prediction for 10 min at 60 A. Claude: put the fitted values into the script defaults in GitHub, upload, `check_sync.sh`.
Pass	RMS error after the fit ≤ 5 K; no constant pinned at a fit range limit; the 60 A prediction is plausible (well below the start-constant value is fine).
If not	RMS above 5 K or constants at limits: keep the old values for those, collect another session (phase 4 + 5).

Phase 8	**Validation and switching on correction**
When	A later ride with a climb.
Setup	Normal riding with stops (brake held 3 s).
Steps	Claude compares model and measurement at each stop in the logs, with the fitted constants. If they agree, Claude sets `therm-meas-correct` to 1 in the script defaults (each stop measurement then resets the rear model), uploads and verifies.
Pass	Model within 5 K of every measurement on a fresh ride.
If not	Leave correction off and repeat the fit with both sessions.

Phase 9	**Summer recheck**
When	First hot day (30–35 °C).
Setup	Same as phase 3–4, without the cooldown.
Steps	Ride the climb with stops; Claude compares model and measurement.
Pass	Model within 5 K; derating thresholds give enough margin.
If not	Refit including the summer logs.

5. Record sheet

Filled in by Claude from the logs and REPL output; Felix adds the room and IR temperatures.

Phase	Date / time	Room °C	FET °C	R (mΩ)	Winding meas. °C	Model °C	IR shell °C	Note

6. Commands used

Command (REPL on CAN 48 via `vesc_cli lisp repl --id 48`)	**Motor action**	**Purpose**
`(therm-test)`	Yes: 25 A, 0.3 s, rear	One measurement: resistance, winding and FET temperature
`(therm-cal)`	Yes: 5 × 25 A, rear	Cold reference R0/T0 into eeprom 100/101
`(therm-cooldown n period)`	Yes: n × 25 A, rear	Cooldown curve, one measurement per period seconds
`(setq therm-meas-interval 20.0)`	No	Denser stop measurements until power-off

Command (REPL on CAN 48 via `vesc_cli lisp repl --id 48`)	**Motor action**	**Purpose**
`(list therm-t1 therm-s1 therm-meas-t therm-derate)`	No	Read model, last measurement, derating
`vesc_cli logs pull`, `tools/fit_thermal.py`	No	Data collection and fit

Why the logs alone are not enough: the back-EMF drifts down about 0.8 % over a ride, which would suggest about 7 K of magnet warming, but the same drift follows the falling pack voltage and cannot be separated from it. The resistance measurements are the ground truth. Accuracy note: the measured resistance includes the cables and controller transistors, which do not warm like copper, so measured rises may read up to about 10 % low; the cold reference taken through the same path keeps this small.